How to Run the Pacman Contest

An exact, step-by-step guide for this MacBook

Team `pacman-agent`

October 2026

The 30-second version

```
cd /Users/bortekstor/Desktop/Eutopia_Pacman_Contest/Pack_It_Up/pacman-agent
source venv/bin/activate
cd pacman-contest/src/contest
python capture.py -r baseline_team -b ../../../my_team.py
```

This is the command from the organisers' `README.md`. Everything below explains each step exactly.

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1 How the files are organised

Path (inside `pacman-agent/`)	What it is
`my_team.py`	**Your team. The only file you edit and submit.** It stays here, at the root.
`TEAM.md`	Your team details (fill in before submitting).
`venv/`	The Python environment (Homebrew Python 3.14 with a working Tk).
`pacman-contest/`	The game engine from the organisers. Do not edit.
`pacman-contest/src/contest/`	Where `capture.py`, `baseline_team.py` and `layouts/` live. You run games from here.
`pacman-contest/agents/`	Old example teams from the engine repository. **Not** where your file goes.

Why `my_team.py` is not in `agents/`

- The organisers' `README.md` says: *"In the root folder ... create in `my_team.py` a class ... "* and runs it as `../../../my_team.py`, which is the root of `pacman-agent`.
- `pacman-contest/agents/team_template/my_team.py` is an outdated example. It uses `import util` instead of `import contest.util`, so it does not load with the current engine.
- In the tournament, the organisers copy your `my_team.py` into their own `teams/<your_team>/` folder. You don't create that folder yourself.

2 Step-by-step: your first game

Step 1 — Open Terminal and go to the project folder

Open **Terminal** (Cmd+Space, type `Terminal`, Enter). Then type:

```
cd /Users/bortekstor/Desktop/Eutopia_Pacman_Contest/Pack_It_Up/pacman-agent
```

Step 2 — Activate the virtual environment

```
source venv/bin/activate
```

What you should see

Your prompt now starts with `(venv)`. Check that the right Python is used:

```
python -c "import tkinter, contest; print(tkinter.TkVersion)"
```

It must print `9.1`. If it prints `8.5` or an error, see Section 7.

Repeat Step 2 in **every new Terminal window**. To leave the environment later, type `deactivate`.

Step 3 — Go to the engine folder

```
cd pacman-contest/src/contest
```

All game commands below are run from **this** folder. Here the short names `baseline_team` and `officeCapture` work, and your team is `../../../my_team.py`.

Step 4 — Watch a game

```
python capture.py -r baseline_team -b ../../../my_team.py
```

What you should see

A game window opens. **Red (left)** is `baseline_team`, **blue (right)** is your `my_team.py`. Each agent gets 300 moves. At the end Terminal prints the result, e.g. `Tie game!` or `The Blue team wins by 5 points.`, and the window closes.

To stop early: click on Terminal and press **Ctrl+C**.

Your `my_team.py` is still the template

Right now `my_team.py` contains the same agents as `baseline_team`, so games usually end in a **tie**. That is expected. Results change once you write your own agents in `my_team.py`.

Step 5 — Fast games without graphics

```
python capture.py -r baseline_team -b ../../../my_team.py -q --delay-step 0
```

What you should see

Within a second: `Red team starts` / `Time is up.` / the result line.

Why `-delay-step 0`? By default the engine sleeps 0.03 s after every move: 36 s of waiting per game, which looks like a hang. The tournament uses `-delay 0.0`, which does the same thing.

3 All useful options

Add any of these to the `python capture.py ...` command (still in `pacman-contest/src/contest`).

Option	What it does
`-l officeCapture`	Use a standard map. Choices: `defaultCapture` (default), `alleyCapture`, `bloxCapture`, `crowdedCapture`, `distantCapture`, `fastCapture`, `jumboCapture`, `mediumCapture`, `officeCapture`, `strategicCapture`, `testCapture`, `tinyCapture`
`-l RANDOM13`	Random map from seed 13 (same seed, same map). Any number works.
`-l RANDOM`	A new random map each game.
`-n 10`	Play 10 games in a row, then print `Scores`, win rates and `Record`.
`--delay-step 0.1`	Slow motion (seconds per move). `0` = full speed.
`-z 0.6`	Window size (smaller than 1 shrinks it).
`-q`	No graphics, minimal output.
`-Q`	No graphics, and hide your agents' own prints.
`-t`	Draw the game as text in Terminal.
`-i 400`	Shorter game: 400 total moves instead of 1200.
`-f`	Fixed random seed, so every run is identical (for debugging).
`--red-name A --blue-name B`	Team names shown in the window.
`--help`	List every option.

3.1 Exact examples

```
# your team as RED instead of blue (the tournament plays both sides)
python capture.py -r ../../../my_team.py -b baseline_team

# office map, slow motion
python capture.py -r baseline_team -b ../../../my_team.py -l officeCapture --delay-step
    0.1

# 10 fast games on random maps, with a summary at the end
python capture.py -r baseline_team -b ../../../my_team.py -l RANDOM -n 10 -q --delay-
    step 0

# your team against itself
python capture.py -r ../../../my_team.py -b ../../../my_team.py
```

Test both sides

In the tournament your team plays as red *and* as blue on random maps. Always test with `-r` and `-b` swapped, and with `-l RANDOM`.

4 Recording and replaying games

Record

```
python capture.py -r baseline_team -b ../../../my_team.py -q --delay-step 0 --
    record
```

Saved as `www/contest_default/replays/match_0.replay` (inside `pacman-contest/src/contest`). Add `-m 7` to save it as `match_7` instead, so the next recording doesn't overwrite it.

Replay with graphics

```
python capture.py --replay www/contest_default/replays/match_0.replay --delay-step
    0.05
```

For tournament replays downloaded from the results page: `python capture.py --replay ~/Downloads/NAME.replay`

The `www/` folder only holds results and recordings. Delete it whenever you like: `rm -rf www`.

5 Playing against your agent yourself

```
python capture.py -r baseline_team -b ../../../my_team.py --keys0
```

You control red agent 0. Click the game window first, then use the **arrow keys** or **W A S D** (**Q** = stop).

6 Checking that it will work in the competition

`CONTEST.md` says the organisers run every match like this:

```
python3 capture.py --delay 0.0 -r teams/<team1>/my_team.py -b teams/<team2>/my_team.py
```

You can reproduce that exactly. From `pacman-contest/src/contest`:

```
mkdir -p /tmp/teams/mine /tmp/teams/other
cp ../../../my_team.py /tmp/teams/mine/my_team.py
cp baseline_team.py   /tmp/teams/other/my_team.py
python3 capture.py --delay 0.0 -r /tmp/teams/mine/my_team.py -b /tmp/teams/other/my_team.
    py
```

If this game runs to the end without an error, your file will load in the tournament.

Checklist before submitting

- ☐ The competition-style test above runs without errors, with your team as red and as blue.
- ☐ No `took too long to make a move` warnings (1 s per move; the third warning loses the game).
- ☐ `create_team()` works with its default arguments. The tournament passes no extra options.
- ☐ No `print(...)` left in `my_team.py`. Everything printed is published with the results.
- ☐ Only Python's standard library plus `numpy`/`scipy`/`sklearn`/`tensorflow`/`keras`, which the organisers guarantee.
- ☐ `TEAM.md` filled in.

Where to submit

The provided files do not say how this contest collects submissions. Follow the organisers' instructions for that step. This folder's git `origin` is the organisers' template (`github.com/aig-upf/pacman-agent`), so you cannot push to it. If they ask for a repository, create your own **private** GitHub repository and point the folder at it:

```
git remote set-url origin git@github.com:<your-user>/pacman-agent.git
git add my_team.py TEAM.md && git commit -m "Submission"
```

```
git push -u origin main
```

7 Troubleshooting

Message / symptom	Cause and exact fix
Window never appears, Python at 100% CPU	Apple's old Python (Tk 8.5) is being used. Run `source venv/bin/activate`. If `TkVersion` still isn't 9.1, rebuild the venv (Section 7.1).
`No module named 'contest'` or `'frozendict'`	venv not active, or packages missing. Activate it, then from `pacman-agent`: `pip install -e pacman-contest frozendict numpy`.
`No teams found!` / `could not be loaded`	You are not in `pacman-contest/src/contest` (Step 3), or the path to `my_team.py` is wrong.
`The layout ... cannot be found`	Same cause: run from `pacman-contest/src/contest`.
Each game takes about 40 seconds	Add `--delay-step 0`.
`KeyError: 'game_id'`	You ran `runner.py`. It is broken in the provided files. Use `python capture.py` as above.
`scripts/play.sh` gives `No teams found!`	It points to `myTeam.py`, which doesn't exist. Don't use it; use the commands above.
`replay.py`: `python3.8` not found	The provided `replay.py` is hard-wired to Python 3.8. Use `python capture.py --replay FILE`.
`Agent 1 took too long to make a move!`	Your agent needed more than 1 s. Make it faster.
`zsh: command not found: python`	venv not active (Step 2).

7.1 Rebuilding the environment from scratch

Only needed if the venv is broken or deleted:

```
cd /Users/bortekstor/Desktop/Eutopia_Pacman_Contest/Pack_It_Up/pacman-agent
deactivate 2>/dev/null; rm -rf venv
/opt/homebrew/bin/python3.14 -m venv venv
source venv/bin/activate
pip install --upgrade pip
pip install -e pacman-contest
pip install frozendict numpy
python -c "import tkinter, contest; print(tkinter.TkVersion)"   # must print 9.1
```

If `/opt/homebrew/bin/python3.14` is missing: `brew install python@3.14 python-tk@3.14`. Do **not** build the venv with plain `python3` (Apple's 3.9 with the broken Tk), and don't run `scripts/config.sh` (it adds the engine from a different repository).

7.2 Updating the game engine

If the organisers announce an engine update, from `pacman-agent`:

```
git submodule update --init --remote
```

```
source venv/bin/activate && pip install -e pacman-contest
```